

The interplay of online attributes and learning modes in shaping satisfaction and effectiveness in online learning

Research Open access Published: 19 November 2024

Volume 3, article number 241 (2024) Cite this article

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Abstract

This study aims to identify key factors influencing the performance and satisfaction of students in online courses during the COVID-19 pandemic. An online survey of 545 Indian university students was conducted, and structural equation modelling was used to analyse the data. The results highlight four critical factors high-quality instruction, well-designed courses, timely feedback, and high expectations that positively impact student satisfaction, which, in turn, enhances performance. These findings underscore the importance of these elements in managing online education effectively. The study offers valuable insights for optimizing online learning environments to boost student satisfaction and overall effectiveness.

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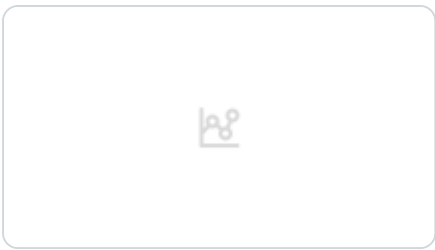
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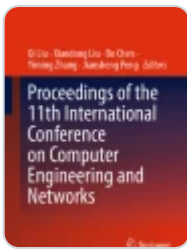
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1 Introduction

Online learning has become increasingly prevalent in the educational system as a result of COVID-19, leading to significant changes in both the student body and the educational landscape. The shift to virtual learning has demonstrated the potential benefits for a wide range of students, providing an engaging and accessible environment. According to the World Economic Forum [64], with careful design and implementation, online learning can enhance the affordability, accessibility, interactivity, and student-centeredness of higher education. Inside Higher Ed's [28] report found that half of the academics surveyed consider online learning an "effective form of teaching," while 87% of online undergraduate and graduate students agreed that it is worthwhile. The impact of COVID-19 is evident in the rise of online learning participation, from 33% in 2017 to 73% in 2020 [27]. Technology continues to play a disruptive yet enabling role in modern education.

This growing reliance on technology in education emphasizes the importance of regional language e-courses, which can drive meaningful advancements in teacher and student training, instructional design, assessment, and planning according to the National Education Policy [NEP], 2020. The focus on technology is further supported by Tata Elxsi's [58] report, which highlights the "4 E's education technology, content, e-learning, and entrepreneurship as key drivers of the next phase of expansion in India's education sector. This expansion is reflected in the emergence of 4450 EdTech companies between 2014 and 2019, showcasing the evolving educational landscape and the growing demand for more effective and satisfying e-learning experiences.

As we delve deeper into the dynamics of online learning, it's essential to distinguish between its different formats. Online learning typically occurs in two main forms: synchronous and asynchronous, with some courses incorporating a blend of both. Research by Perveen [50] suggests that the mode of learning whether synchronous, asynchronous, or mixed significantly influences the effectiveness and satisfaction of e-learning. According to the [65] with careful design and implementation, online learning can enhance the affordability, accessibility, interactivity, and student-centeredness of higher education. Inside Higher Ed's [28] report found that half of the academics surveyed consider online learning an "effective form of teaching," while 87% of online undergraduate and graduate students agreed that

it is worthwhile. The focus on technology is further supported by Tata Elxsi's [58] report, which highlights the "4 E's" education technology, content, e-learning, and entrepreneurship as key drivers of the next phase of expansion in India's education sector. This review explores the impact of these learning modes on user satisfaction, overall effectiveness, and the quality of learning. The study also includes recommendations for future research and practical applications, informed by a thorough review of relevant literature, materials, methodologies, findings, and conclusions.

1.1 Theoretical development and hypotheses formulation

1.1.1 Influence of learning attributes on satisfaction

Any learning that uses the internet to give some sort of education to a student or learners separated by time, distance, or both is referred to as online learning [12]. The effectiveness and accessibility of the technology, which includes connectivity even at considerable distances, enhanced convenience, and expanded educational options, have been the focus of numerous studies on online learning [13, 32, 40, 41, 49]. Students give course design and implementation priority, according to Singleton et al. [55]. The acceptability of online learning could be increased by better models and processes. Important characteristics that have a big impact on learner satisfaction are efficiency and convenience of use [2]. [14] found three satisfaction components: engaged learning, agency, and assessment. Students Favor active over passive learning, expecting high interactivity in classes [15]. [45] found students' use of the audio feature significantly predicted their class involvement. [5] concludes with high-impact principles for online education, emphasizing instructional design, effective delivery, support, participation, and contingency plans.

[14] identified six key components that contribute to student satisfaction in online learning: an enriched learning environment, instructor commitment, learner engagement, collaborative learning opportunities, timely feedback, and ease of access to resources. These elements are crucial for creating a supportive and engaging learning atmosphere, enhancing student motivation, and ensuring effective course navigation. [30] highlighted five essential elements for satisfaction: perceived usefulness, ease of use, content quality, interaction, and service quality. These factors directly impact the usability and effectiveness of online platforms, ensuring that technology is intuitive, valuable, and academically rigorous while fostering interaction and adequate support. [23] emphasized the importance of interactive learning quantity, cultural relevance, and the nature of the program. Interactive learning quantity ensures high engagement levels, cultural relevance makes the content more inclusive, and the program's nature must align with student expectations to maintain satisfaction. According to [35], student evaluations are positively influenced by the efficient use of technology and engaging interactions between learners and instructors, as well as meaningful learner-content interactions. Effective technology enhances accessibility and interactivity, while engaging interactions contribute to a supportive learning environment and deeper understanding of the material.

Effective technology solutions for online courses are increasingly designed with a keen awareness of student expectations [34]. Recent research continues to underscore that the effectiveness and convenience of e-learning tools play a critical role in student satisfaction. Meeting student expectations not only enhances satisfaction but also positively influences their overall learning experience [1, 57]. This aligns with updated models of student satisfaction, such as the "Online Learning Satisfaction Model" [67]. Conversely, unmet expectations can lead to reduced course satisfaction and negatively impact learning outcomes.

Thus, we hypothesize,

H1

Specific Attributes of Online Learning Influence Student Satisfaction in Online Courses.

1.1.2 Satisfaction and effectiveness of learning [11]

Emphasize the importance of high-quality online resources, faculty interactions, student-friendly software, multimedia elements, and a conducive home environment for successful online learning. They also note the significance of adjustable sound levels and PowerPoint presentations. [63] identifies three crucial factors for online learning effectiveness: a preference for virtual instruction, proficiency in digital communication tools and internet use, and the ability for self-directed education. Effective online learning depends on interactive instruction, consistent course design [4, 40, 42], and instructors fostering critical thinking [18, 21]. Interaction rate [43, 54] and flexibility [31] are crucial. Technological skills remain vital [41, 51]. Well-structured courses, knowledgeable instructors, technology, feedback, and clear instructions are essential for success [47, 57]. [19] found that a good instructor, well-designed courses, prompt feedback, and high expectations positively impact satisfaction and student performance. High-quality instructors adapt to student's needs [37]. Prompt feedback strengthens teacher-student bonds, improving learning outcomes [8, 55]. [27] examines the impact of instructor quality, course design, and learner expectations on satisfaction in online learning. Data from 121 students in a Tourism Management program showed all three factors significantly influence learner satisfaction. The findings highlight the importance of these elements in enhancing the online learning experience and suggest exploring additional factors, such as demographics, in future research.

In this context, we hypothesize,

H2

Satisfaction with online learning positively correlates with its effectiveness.

H3

Online learning attributes influence the effectiveness of online learning.

1.1.3 Influence of mode of learning on effectiveness

According to [51], there are now several types of online learning environments: synchronous, asynchronous, hybrid, and more recently, fully remote and blended learning, which have become prominent due to the COVID-19 pandemic. Real-time cooperation and instructor-led activities with simultaneous attendance are features of synchronous sessions [53]. Self-paced learning using recorded lectures is flexible in asynchronous situations [22]. According to [24], hybrid learning environments combine synchronous meetings with asynchronous activities, offering a balance between flexibility and real-time interaction.

Clear design, open contact with instructors, and active involvement in class are critical for student happiness and perceived learning in asynchronous mode, according to [25]. It is essential to comprehend how kids view these two learning activities [44, 56]. Asynchronous learning environments are student-centered because they allow for self-directed learning [46]. Asynchronous online learning fosters critical thinking and in-depth learning through peer discussions, adding new concepts to what has already been learned [25, 36].

Participating in remote learning fosters a low affective filter by reducing shyness and instructor anxiety. Asynchronous learning reduces pressure, enabling students to respond quickly and creatively. Students using asynchronous computer-mediated communication (CMC) showed greater discourse functions linked to question types and techniques than their synchronous counterparts, according to studies by [3, 36]. While asynchronous CMC encourages asking questions that require in-depth, lengthy responses, clarification, and extension, synchronous CMC encourages simple, succinct answers [26, 29]. While 75% of students indicated a preference for synchronous lectures, recent research by [38] found that both synchronous and asynchronous course delivery methods can be

equally effective in achieving learning outcomes. Asynchronous online learning, according to [24], allows students to combine their academic obligations with those of their families, jobs, and studies, which results in more considered contributions. According to [66], synchronous meetings strengthen student relationships and participation. In synchronous situations, providing real-time feedback improves community development and student engagement [62, 66].

Considering this, we propose the hypothesis,

H4

Mode of learning has a moderating effect between online learning attributes and satisfaction with online learning.

2 Methodology

A structured online survey was conducted for this cross-sectional study, utilizing judgment sampling to select participants. The study aimed to gather insights from a diverse demographic, including male and female students from various age groups and both undergraduate and graduate programs in hotel management and management studies. Informed about the study's purpose and confidentiality, participants were not offered incentives. The survey, created using Google Forms, was distributed via email to 15 colleges in India during the COVID-19 lockdown, yielding 545 responses. This sampling method was essential for capturing the behaviours of respondents engaged in both synchronous and asynchronous online learning, ensuring representation from rural and urban areas while minimizing bias. Care was taken to include a broad spectrum of educational and social backgrounds for a comprehensive understanding of learner’s perceptions.

2.1 Data analysis and results

To analyze the data, both SPSS and AMOS software were employed. An exploratory factor analysis (EFA) with VARIMAX rotation was conducted on a sample of 545 respondents, including 54% females with ages ranging from 15 to 50. Research indicates that 73% use smartphones, 21% use laptops, 5% use tablets, and 1% use other gadgets for online learning. Of the total respondents, 368 utilized a synchronous online system, while 177 opted for asynchronous learning. Approximately 45% of respondents are comfortable with online learning, yet many report distractions at home. The main disruptions are from colleagues, though 59% did not face any disturbances. Most prefer synchronous learning with the camera and mic off, while 32% Favor asynchronous learning. Moderation effects were assessed by comparing critical ratio values and regression values between the two groups. To assess the degree of moderation effect and to test the overall validity of the hypothetical model, Excel statistics were used to compare the regression weights, and P values and Z scores were used to assess the degree of moderation effect.

In Table 1, the majority of students (56.1%) preferred moderate-sized groups of up to 10 students for online learning. About one-third preferred smaller two-person teams. Very large groups of over 10 students were least favourable, preferred by only 12.8% of students. In summary, small to medium-sized online learning groups were most widely preferred, while large groups greater than 10 were less ideal for most students.

Table 1 Preferred group size for online learning

It is observed, that nearly half of students (45.3%) were comfortably able to learn from home. However, 29.7% reported getting easily distracted by other home activities. Some faced situational challenges like interruptions (13.4%) or unsuitable home conditions (11.6%). In summary, while close to half of students could learn comfortably online at home, over 50% faced some type of issue like distractions, interruptions, or suboptimal environments interfering with their remote digital learning. Supplementary support in these areas may help further facilitate effective home online learning for students (see Table [2](#)).

Table 2 Experience with online learning from home digitally

Over half of students (52.8%) find video lectures by their faculty most effective, emphasizing the impact of instructor familiarity. Industry expert videos are also well-received (27.7%), while lectures from overseas universities (11.2%) or unknown experts (8.3%) are less favored. In essence, students prioritize familiarity with online learning videos, underscoring the significance of student–teacher connections (see Table [3](#)).

Table 3 Type of recorded video lecture more effective for learning

The data shows that online multiple-choice quizzes were the most common among the students surveyed, with 42.6% taking this type of quiz. This suggests that during the COVID-19 pandemic, many courses and assessments shifted online in the form of multiple-choice questions. Traditional quizzes still comprised a significant portion too. when combined, traditional multiple choice and short answer formats made up 47.2% collectively. However, the online shift is evident based on online quizzes representing over 50% of the total. The least common type of quiz was the online short answer format. This indicates that while assessments moved online, they tended to keep using simpler multiple-choice questions rather than open-ended short responses which may be more difficult to administer and grade virtually (see Table [4](#)).

Table 4 Type of quiz more effective for testing knowledge

This data represent clear visibility of presentation slides (38.7%) was deemed the most effective, followed by audio clarity (24.6%) and reduced commuting (20.6%). Surprisingly, only 16.1% valued the ability to ask questions without reservation, indicating a need for improved interactivity in online formats. In summary, content visibility and convenience are highly valued in online learning, but there's room for improvement in enhancing interactive elements for a more engaging experience. According to online learners, the major effective attribute of online learning is the availability of PPTs right in front of every learner followed by the ability to hear the lecture (see Table [5](#)).

Table 5 Effective attributes for online teaching learning

Most students (59.3%) reported no disturbances during online learning at home. However, 31.6% experienced occasional interruptions from colleagues, and 9.2% faced constant disruptions. Overall, 40.7% encountered some level of disturbance from home occupants, suggesting a need to improve online learning spaces for a more focused study environment in shared living situations (see Table [6](#)).

Table 6 Types of disturbances at home for online learning

The majority (61.3%) had moderate home duties, finding ample time for online classes. About 25.7% had minimal obligations, while 13% faced time constraints due to numerous responsibilities. While most balanced life duties, over 10% faced challenges, emphasizing the need for flexible scheduling in online education (see Table [7](#)).

Table 7 Responsibilities at home and time for online learning

Most students (60.6%) prefer directly asking teachers to resolve doubts, emphasizing the importance of real-time interaction. Reviewing class materials (22.8%) and consulting peer discussion forums (16.7%) were secondary strategies. Despite the shift online, students overwhelmingly value instant clarification from professors, highlighting the importance of robust instructor availability for effective doubt resolution (see Table [8](#)).

Table 8 A most preferred method for clearing doubts

The majority (73.6%) used smartphones for remote learning, prioritizing accessibility and convenience during COVID restrictions. Few relied on laptops/desktops (21.1%), tablets (4.6%), or other gadgets (0.7%). This highlights the importance of mobility and ease of access over advanced features in online engagement, showing the pivotal role of smartphones in learning from anywhere (see Table [9](#)).

Table 9 Mostly used devices for online learning

The top preference (43.3%) was synchronous learning with the camera and microphone off, indicating a desire for real-time lessons while maintaining privacy. The second choice (24.2%) was synchronous with partial microphone participation. Purely self-paced asynchronous options were less favored (32.4%), highlighting a preference for live instruction with muted involvement, allowing for connection without revealing potentially distracting home environments (see Table [10](#)).

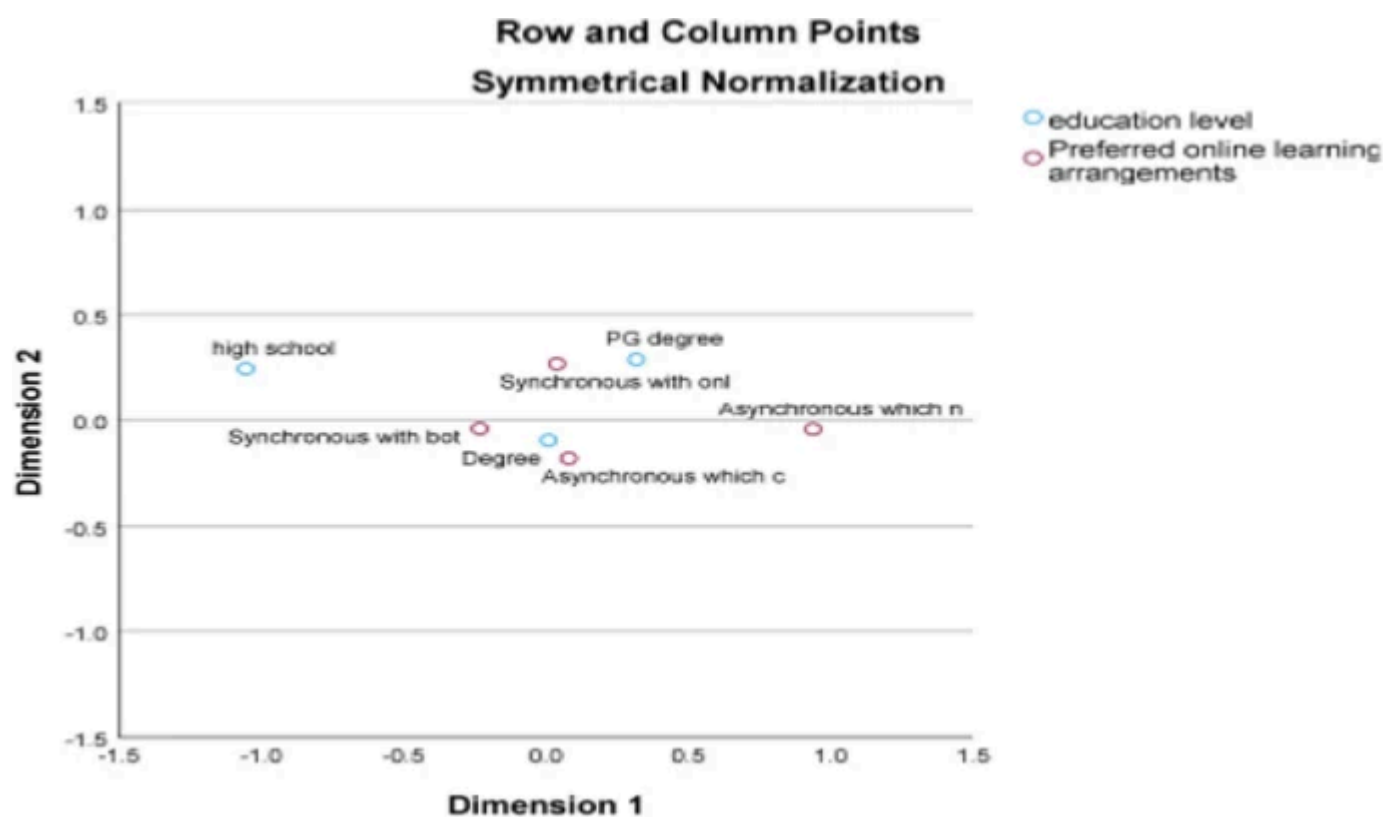
Table 10 Preferred online learning mode

In Table 11 majority (41.5%) of preferred keeping cameras off unless interacting, with 32.8% remaining neutral. Only 25.7% wanted cameras on continually. Overall, 74.3% either preferred intermittent camera usage or had no strong preference, suggesting persistent video feeds were unnecessary for over two-thirds of students. The data indicates that always-on cameras provide little benefit compared to targeted activation upon request.

Table 11 Preference for camera usage–switch on\off

The interrelationship between respondents' various positions on the usage of synchronous and asynchronous learning modes, based on their education levels, is illustrated through correspondence analysis (Fig. 1; Tables 12, 13). It was found that postgraduate students generally prefer synchronous learning with the microphone switched off. In contrast, graduates (those who have completed their degrees but are not yet postgraduate) tend to Favor synchronous sessions with both the microphone and camera off or prefer asynchronous modes that allow for flexible viewing at their convenience. This pattern indicates that students' seriousness significantly influences their preferences for learning modes.

Fig. 1



Symmetrical representation (correspondence)

Table 12 Correspondence table (usage of synchronous and asynchronous learning modes)

Table 13 Correspondence table (summary)

2.1.1 Measurement model

Firstly, the gathered data underwent scrutiny for missing values and outliers, revealing no issues in either category. Subsequently, the skewness and kurtosis of study items were examined to assess univariate normality. Results indicated that responses to all items were reasonably free from skewness and kurtosis, with standardized values outside the range of 1.96. This analysis aligns with standards set by [6, 10, 20, 33, 48], indirectly supporting the assumption of multivariate normality.

In the study, Confirmatory Factor Analysis (CFA) was employed to analyze the measurement model and validate constructs and measurement properties after confirming necessary Structural Equation Modelling (SEM) assumptions (Fig. 2). Beyond the typical Chi-square values, goodness-of-fit indices, including CFI, IFI, RMSEA, and SRMR, were used to assess the measurement model's fit. Results for learning attributes (CFI = 0.99, IFI = 0.99, RMR = 0.015, RMSEA = 0.066) and learning effectiveness (CFI = 0.98, IFI = 0.98, RMR = 0.026, RMSEA = 0.072) supported a good fit. The study validated dimensions according to Netemeyer et al. [47] for validity and reliability. Composite Reliability (CR) values exceeded the recommended cutoff of 0.80, and Average Variance Extracted (AVE) values surpassed the specified cutoff of 0.50, confirming convergence validity and reliability. Discriminant validity was tested by analyzing the square of each pair of correlations with AVE values, and the findings consistently demonstrated that AVE values were higher than the correlation square between the pairs, supporting discriminant validity.



Structural equation modelling testing the relationship between the Online attribute preferences and the learning effectiveness.

2.1.2 Hypotheses testing:testing

The study's main objective was to employ Structural Equation Modelling to assess the given set of hypotheses after the measurement model had been verified. According to the structural model results (Table 15), the model fits the data quite well ($\chi^2 = 335.75$, $df = 120$, CFI = 0.97, IFI = 0.97, RMR = 0.053, RMSEA = 0.042). All the hypotheses are tested valid at a 5 percent level of significance. The model fit indices are all within the acceptable range. (see Table 14).

Table 14 Hypotheses

The effect of moderation was tested with two sub-models wide Synchronous and Asynchronous learning modes. The comparison of regression estimates, critical ratios, and P values are shown below in Tables [15](#) and [16](#).

Table 15 Regression weights: (synchronous—default model)

Table 16 Regression weights: (asynchronous—default model)

It was found that the p-values corresponding to each variable relationship are significant except for the direct relationship between online attributes and the effectiveness of online learning in the asynchronous sub-model. Since the Z-score values for checking the significance of group-wise differences between regression estimates of two paths are not within the range between -1.96 and + 1.96 (i.e., Satisfaction causes online effectiveness in the asynchronous learning mode and the effect of Web-based traits on online effectiveness in synchronous learning mode). It is deduced from this study that there is an effect of moderation due to the type of learning mode in the association between satisfaction and online effectiveness, particularly in the asynchronous learning mode, and a significant effect of moderation in the direct path between Web-based traits and effectiveness in the synchronous learning mode (Table [17](#)). It is observed from Tables [16](#) and [17](#) that mere attributes did not create any direct positive result on online learning in the case of asynchronous mode of learning.

Table 17 Test of moderation effect on the effectiveness of online learning caused by learning mode type

Online learners undergoing degree courses are more tempted to use synchronous learning mode with both camera and mike switched off whereas postgraduates have tendencies to follow synchronous (live class with only mike switched off) (Table [10](#)).

As shown in Table [17](#), It is clear from the regression weights that the influence of digital attributes on the effectiveness of online learning gets strengthened if the students are satisfied with the online attributes. This shows a mediating effect of satisfaction in the association between web-based traits and the effectiveness of learning. Moreover, the mode of learning, particularly the asynchronous type has a greater moderation effect in strengthening the association between satisfaction and Learning proficiency. This means online learning effectiveness increases on account of learners' satisfaction with online features if the asynchronous mode is used by the learners.

3 Discussion and implications

The results of this study reveal that flexibility in online learning environments plays a pivotal role in enhancing student satisfaction, which in turn positively influences overall Learning proficiency. Specifically, the data showed that when students are given the option to attend sessions based on their individual preferences, whether synchronously or asynchronously, their engagement and satisfaction significantly increase [[9](#), [13](#), [61](#)]. This finding is particularly noteworthy, as it suggests that the effectiveness of asynchronous learning does not always translate into positive learning outcomes. Unlike synchronous sessions, which allow for real-time interaction and immediate

feedback, asynchronous formats may lead to feelings of isolation or disconnection among students if not carefully designed. Previous research supports this perspective, indicating that asynchronous learning environments can lack the engagement factors found in traditional classroom settings, which may hinder Learning proficiency (e.g. [68]). Therefore, a deeper investigation into the specific elements of asynchronous learning that contribute to student satisfaction is warranted.

In addition to flexibility, the study highlights the significant impact of platform usability, appeal, and efficiency on student satisfaction, particularly in asynchronous settings. The results indicate that online platforms must be intuitive, accessible, and mobile-friendly to ensure that students from diverse backgrounds can easily navigate them [52, 60]. These findings are especially relevant for educational institutions aiming to create inclusive learning environments that accommodate students with disabilities and those facing technical challenges such as poor internet connectivity [7]. Finding is related to offer valuable insights into the effects of various teaching modalities on student satisfaction and learning outcomes. They provide educators with important guidance for refining instructional strategies, helping to improve student learning efficacy across different formats. The implications of these results suggest that institutions should invest in both user-friendly platforms and the necessary infrastructure to provide equitable access to online education for all students, including those in underserved regions [17, 39]. Ensuring these technological advancements will be critical in enhancing the accessibility and quality of online learning experiences. For example, the integration of adaptive technologies and personalized learning paths can address individual student needs, which is increasingly vital in today's diverse educational landscape.

Moreover, the study's results emphasize the importance of continuous professional development for educators and the strategic use of learning analytics. Instructors must be trained to effectively utilize educational technologies and adapt their teaching strategies based on real-time feedback and student engagement metrics [25, 62]. This approach aligns with findings from other studies indicating that ongoing professional development significantly enhances educators' capacity to engage students and leverage technology effectively [30, 36]. By applying data-driven insights, institutions can refine their course offerings to better meet the needs of their students. These implications align with the broader goals of Sustainable Development Goal 4 (SDG 4), which advocates for inclusive and equitable quality education, particularly in light of the COVID-19 pandemic's acceleration of online learning [16, 59]. By addressing the findings of this study, policymakers and institutions can take actionable steps to improve the design and delivery of online education, ensuring that it remains effective and accessible to a diverse range of learners.

4 Scope for further research

The scope for future research in this study opens several avenues for expanding the understanding of online learning's effectiveness, particularly in light of the current limitations. One critical area for further exploration is the reliance on self-reported data, which introduces potential bias. Students' satisfaction and perceptions of learning may fluctuate due to mood, external pressures, or transient circumstances rather than the actual quality of the learning environment. Future studies could incorporate more objective measures, such as performance metrics or behavioural data (e.g., engagement analytics), to complement subjective responses and offer a more robust understanding of Learning proficiency. Additionally, longitudinal studies tracking the long-term impact of learning modalities on student performance could provide deeper insights into how student satisfaction translates into academic success over time.

Another significant research opportunity lies in broadening the demographic and geographic scope of studies on online learning. Since this research focused exclusively on students in India, the findings may not fully represent global online education dynamics. Future research could involve cross-cultural comparisons or studies involving

participants from diverse educational systems to explore how cultural, social, and economic factors impact online learning satisfaction and performance. By examining students from different countries or regions, researchers can investigate the extent to which variables like technological infrastructure, cultural attitudes toward education, and varying access to educational resources influence the success of online learning environments.

Finally, expanding the range of factors analyzed in relation to student satisfaction and Learning proficiency presents a compelling direction for future research. This study narrowed its focus to four key attributes usability, appeal, usefulness, and efficiency but there are numerous other psychological, social, and environmental variables that likely contribute to online learning success. Future studies could explore the impact of factors such as motivation, self-regulation, cognitive load, and the physical learning environment. Additionally, examining demographic variables such as age, gender, socioeconomic background, and prior experience with online learning could yield a more comprehensive understanding of how these characteristics influence online education outcomes. Incorporating these broader variables would enrich the knowledge base and provide more actionable insights for educators and policymakers aiming to improve online learning systems.

5 Conclusion

Online learning has become a way of life among learners' communities, especially in the aftermath of COVID-19. Furthermore, many professionals explored and realized the importance of upskilling through online platforms. This study was an attempt to understand the role of learning attributes in the effectiveness and satisfaction of online learning. Additionally, the study explored the moderating role of the mode of learning. It was found that learning attributes and modes of learning do play a crucial role in the effectiveness of online learning. The study underscored the importance of technical features, the trainer's role, students' engagement, and environmental settings which contribute to the effectiveness and subsequent satisfaction of online learning. In the wake of the increased prevalence of EdTech and e-learning platforms, studies in this direction largely contribute to the enhancement of quality online learning and the development of tech ventures.

Data availability

We had gathered data by using a Google Form. The data excel sheet has been uploaded to the supplemental material. We made sure that participants' welfare and adherence to ethical standards were maintained throughout the study by following ethical guidelines. This Study is conducted in higher education where students are above 18 years of age.

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Acknowledgements

Not applicable. No external support received.

Funding

No funding support was received for this research work.

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Contributions

Dr. Paul, as the lead author of this publication, coordinated the entire process of the article. Dr. Jomon played a pivotal role in conceptualizing the paper and conducting data analysis, contributing significantly to the development of the model and the overall article. Mr. Kushwaha assisted with the documentation.

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Ethics declarations

Ethics approval and consent to participate

The study was conducted in accordance with the international ethical standards outlined in the Helsinki Declaration by the World Medical Association. All standard practices for minimizing risk to participants were strictly followed, including ethical guidelines for data collection. All participants voluntarily provided informed consent to participate in the study. This research did not involve any medical, biological, personal, or privacy-sensitive elements and, as per the guidelines of XIME University Bangalore, did not require an ethics approval statement from the Institutional

Ethics Committee. However, all relevant ethical standards were upheld throughout the study. The research was conducted under the umbrella of XIME Kochi, Kerala, India. Necessary permissions were obtained from the Cochin International Airport Authority, Government of Kerala, and the Ministry of Aviation, Government of India to conduct this study. The manuscript is submitted for publication in compliance with the journal’s ethical requirements.

Consent for publication

Informed consent was obtained from all participants involved in the study. Participants were informed about the potential research results, and consent was provided for the use of collected data. Additionally, consent to publish was granted by all participants. In cases where participants were under the age of 18, consent was obtained from a parent and/or legal guardian.

Competing interests

The authors declare no competing interests.

Additional information

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Supplementary Information

Below is the link to the electronic supplementary material.

[Supplementary file1 \(CSV 567 KB\) \(download CSV ↴\)](#)

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Cite this article

Mathew, P.V., Pappachan, J. & Kushwaha, S. The interplay of online attributes and learning modes in shaping satisfaction and effectiveness in online learning. *Discov Educ* 3, 241 (2024). <https://doi.org/10.1007/s44217-024-00344-y>

Received
08 February 2024

Accepted
06 November 2024

Published
19 November 2024

Version of record
19 November 2024

DOI
<https://doi.org/10.1007/s44217-024-00344-y>

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Keywords

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